

Course- **B. Tech**
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Course Name- **AI in Healthcare**
Semester- **VII**

Lab Experiment 1:

Objective: Implementation of program to read tabular, textual, image, signal, and medical data in different formats.

Detailed Explanation: To Read Tabular, Textual, Image, Signal, and Medical Data in Different Formats

A robust data ingestion module for multimodal scientific/medical research must handle diverse data types, each with its own file formats, metadata conventions, and preprocessing needs. Below is a breakdown of the modalities and what a generalized implementation should support:

A. TABULAR DATA

Common formats: CSV, Excel (XLS/XLSX), JSON, SQL exports,.

Key steps:

- ✓ Automatic detection of delimiter/encoding.
- ✓ Schema inference (data types, missingness).
- ✓ Handling categorical vs numerical features.
- ✓ Parsing dates/timestamps.
- ✓ Validation (ranges, constraints).

Libraries (Python): pandas for CSV/Excel/JSON, sqlalchemy for relational imports.

B. TEXTUAL DATA

Common formats: Plain text (.txt), structured (JSON/XML), clinical notes (sometimes in rich text or semi-structured), PDFs, HTML.

Key steps:

- ✓ Extraction (e.g., PDF → text with pdfplumber or PyPDF2).
- ✓ Cleaning (removing headers/footers, de-identification if needed).
- ✓ Tokenization, stopword removal, basic NLP features (term frequency, embeddings).
- ✓ Metadata association (e.g., which patient/note/date).

Libraries: nltk, spacy, transformers (for embeddings), pdfplumber or PyMuPDF.

C. IMAGE DATA

Common formats: PNG/JPEG, DICOM (medical), TIFF, NIfTI (neuro), etc.

Key steps:

- ✓ Reading and normalizing pixel values.
- ✓ Handling varied resolutions and channels.
- ✓ Extracting metadata (e.g., from DICOM headers).
- ✓ Preprocessing: resizing, windowing (medical), augmentation if needed.

Libraries: Pillow, opencv-python, pydicom for DICOM, nibabel for NIfTI.

D. SIGNAL DATA

Examples: ECG, EEG, respiratory waveforms.

Formats: WFDB (PhysioNet), EDF, plain CSV with timestamps.

Key steps:

- ✓ Sample rate detection.
- ✓ Filtering (e.g., bandpass to remove noise).
- ✓ Annotation alignment (e.g., R-peaks in ECG).
- ✓ Segmentation (beats, epochs).

Libraries: wfdb / wfdb-python, scipy.signal.

E. MEDICAL DATA (COMPOSITE)

Often a combination: clinical tabular (labs, demographics), imaging, signals, and free-text notes.

Challenges:

- ✓ Patient linkage across modalities (unique IDs).
- ✓ Time alignment.
- ✓ Privacy / de-identification.
- ✓ Heterogeneous missingness.

Design of the Program:

Modular pipeline with the following components:

1. **Loader modules** per modality (tabular_loader, text_loader, image_loader, signal_loader).

2. **Validator/Inspector** to summarize each loaded object (shapes, sample statistics, missing values).
3. **Preprocessor** to standardize (normalize images/signals, encode categories, clean text).
4. **Fusion interface** (optional) to align modalities by subject/time for downstream modeling.
5. **Export/logging** for reproducibility (e.g., manifest of sources, versions, checksums).

A simplified driver in Python would detect file types, dispatch to appropriate loaders, and accumulate a unified Dataset object with metadata.

DATASET SELECTION AND DETAILED DISCUSSION

1. Primary Tabular & Medical Dataset: UCI Cleveland Heart Disease Dataset

Source: [UCI Machine Learning Repository](#). This is the canonical “Cleveland” subset used in most published experiments.

- [Heart Disease Dataset](#)

Content: 303 patient records with originally 76 attributes; standard practice uses a subset of 14 relevant features for predicting presence of heart disease (binary classification: disease vs no disease). Attributes include age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, resting ECG results, maximum heart rate achieved, exercise-induced angina, ST depression, and others.

Feature description reference:

(https://www.researchgate.net/publication/354492312_Heart_Disease_Risk_Prediction_Using_Machine_Learning_Classifiers_with_Attribute_Evaluators/figures?lo=1&utm_source=chatgpt.com&utm_medium=&utm_campaign=)

Medical relevance: Used extensively in cardiovascular risk prediction research, e.g., logistic regression, neural nets, fuzzy systems, etc. [PMC](#)

Format: Available as CSV; easily ingestible via pandas.

2. Signal Dataset: MIT-BIH Arrhythmia Database (ECG Signals)

Source: PhysioNet’s MIT-BIH Arrhythmia Database—standard benchmark for ECG signal analysis.

- <https://www.physionet.org/content/wfdb/10.7.0/>
- <https://www.physionet.org/content/mitdb/1.0.0/>

Content: 48 half-hour two-channel ambulatory ECG recordings from 47 subjects, with expert annotations for arrhythmias. Includes beat-type labels used for rhythm classification.

Reference- title” Verification and comparison of MIT-BIH arrhythmia database based on number of beats.”

Access & Tools: The wfdb-python package provides native reading, visualization, and annotation handling. physionet.org/WFDB

Format complexity: Files come in .dat, .hea, .atr—interpreted via WFDB APIs. Tutorials and examples (including Colab-compatible demos) are available. [MediumStack Overflow](#)

IMPLEMENTATION PLAN FOR GOOGLE COLAB

The Colab notebook will perform the following:

1. Environment setup

- Install required packages: pandas, scikit-learn, wfdb, matplotlib, seaborn (optional for visualization if allowed), numpy.

2. Tabular data ingestion (Heart Disease)

- Download Cleveland Heart Disease CSV from UCI (or Kaggle mirror).
- Inspect schema, handle missing values, map target to binary (0 vs 1+).
- Basic exploratory data analysis: distributions, correlations.
- Train a baseline classifier (e.g., logistic regression) to predict heart disease presence.
- Evaluate with accuracy, ROC AUC, confusion matrix.

3. Signal data ingestion (ECG)

- Use wfdb to fetch a sample MIT-BIH record (e.g., record “100”).
- Load the signal and annotation.
- Plot a segment of ECG waveform with beat annotations.
- Compute basic heart-rate variability or beat interval statistics.

4. Textual data demonstration

- Include a synthetic or small sample clinical note (e.g., a plain text “patient note” describing chest pain symptoms).
- Tokenize, compute word frequency, and extract key terms (simple NLP).

5. Image data demonstration (optional placeholder)

- Use a publicly available medical image (e.g., a sample chest X-ray from a CC0 collection) or synthetic image.
- Load with Pillow, display, and resize.
- Extract metadata if possible.

6. Integration Discussion

- Outline (in markdown) how heart disease risk (tabular) could be augmented with ECG-derived features, and how text notes could provide context (symptoms) in a future fused model.